# General Mathematics · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

**Name** _____ **Date** _____

| Exam 1 · multiple choice | Exam 2 · written response | State average on these questions |
|---|---|---|
| **40 marks** | **60 marks** | **72 / 100 (72%)** |
| reading 15 min · writing 1 h 30 min | reading 15 min · writing 1 h 30 min | a student at the state average on every question |

**How to use this paper**

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright.
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Working can be typed in plain text, e.g. 0.004 × 5440.88 = 21.76.
3. Keep to the real time limits above, with your CAS, your bound reference and the VCAA formula sheet, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

**Why these questions**

Every question is a real VCAA exam question from the 2023–2025 General Mathematics exams, chosen by ranking all the questions by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions) within each area of study. This paper takes the highest band, with the real exam's balance of areas.

**Sources**

- 2023: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2024: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2025: Exam 1 (PDF) · report · Exam 2 (PDF) · report

## Exam 1 · multiple choice (40 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then click your answer. 2023 questions have five options (A–E); 2024 and 2025 questions have four (A–D).

### Data analysis

**Question 1 · 1 mark**

2023 VCAA Exam 1 · Question 1 · open in the official exam PDF (p. 3) ↗

Median from a dot plot of 40 runners' 800 m times

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 2 · 1 mark**

2023 VCAA Exam 1 · Question 2 · open in the official exam PDF (p. 3) ↗

Describe the shape of the same dot plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 3 · 1 mark**

2023 VCAA Exam 1 · Question 3 · open in the official exam PDF (p. 4) ↗

Sixth value needed to make the mean of six puzzle attempt counts equal 8

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 4 · 1 mark**

2023 VCAA Exam 1 · Question 4 · open in the official exam PDF (p. 4) ↗

Expected number of 2380 museum visitors within two standard deviations of the mean visit time

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 5 · 1 mark**

2023 VCAA Exam 1 · Question 5 · open in the official exam PDF (p. 4) ↗

Height from a z-score of −0.85, given the mean and standard deviation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 6 · 1 mark**

2024 VCAA Exam 1 · Question 2 · open in the official exam PDF (p. 4) ↗

Types of two survey variables (payment preference and budget preference)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 7 · 1 mark**

2024 VCAA Exam 1 · Question 5 · open in the official exam PDF (p. 5) ↗

Which statement about a boxplot of 24 sibling counts is correct

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 8 · 1 mark**

2024 VCAA Exam 1 · Question 7 · open in the official exam PDF (p. 6) ↗

Standard deviation from a mean, a score and its standardised score

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 9 · 1 mark**

2025 VCAA Exam 1 · Question 3 · open in the official exam PDF (p. 3) ↗

Which statement comparing two parallel boxplots of life expectancy is correct

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 10 · 1 mark**

2025 VCAA Exam 1 · Question 5 · open in the official exam PDF (p. 4) ↗

Mean and standard deviation from two percentages of a normal distribution

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 11 · 1 mark**

2025 VCAA Exam 1 · Question 6 · open in the official exam PDF (p. 5) ↗

Number of lower outliers given $Q_1$, $Q_3$ and the five lowest values

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 12 · 1 mark**

2025 VCAA Exam 1 · Question 7 · open in the official exam PDF (p. 5) ↗

Percentage of female buyers preferring silver, from a two-way table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 13 · 1 mark**

2025 VCAA Exam 1 · Question 8 · open in the official exam PDF (p. 5) ↗

Best graph for a two-way table of colour preference by sex

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 14 · 1 mark**

2025 VCAA Exam 1 · Question 11 · open in the official exam PDF (p. 7) ↗

Least squares line with $\log_{10}$(life) as the response and doctors as the explanatory variable

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 15 · 1 mark**

2025 VCAA Exam 1 · Question 14 · open in the official exam PDF (p. 8) ↗

Qualitative features of the margin time series

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 16 · 1 mark**

2025 VCAA Exam 1 · Question 15 · open in the official exam PDF (p. 9) ↗

Four-mean smoothed value with centring for day 8 of juice bar sales

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

## Recursion and financial modelling

**Question 17 · 1 mark**

2023 VCAA Exam 1 · Question 18 · open in the official exam PDF (p. 12) ↗

Rule for the value after n years from a unit cost depreciation recurrence relation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 18 · 1 mark**

2023 VCAA Exam 1 · Question 19 · open in the official exam PDF (p. 12) ↗

Number of cups made per year from the yearly depreciation and cost per cup

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 19 · 1 mark**

2023 VCAA Exam 1 · Question 20 · open in the official exam PDF (p. 13) ↗

Flat rate depreciation rate for a $3000 computer worth $600 after four years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 20 · 1 mark**

2024 VCAA Exam 1 · Question 18 · open in the official exam PDF (p. 11) ↗

Annual interest rate from a weekly loan recurrence relation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 21 · 1 mark**

2024 VCAA Exam 1 · Question 19 · open in the official exam PDF (p. 11) ↗

Value of k in a reducing balance recurrence relation for 18% per annum

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 22 · 1 mark**

2024 VCAA Exam 1 · Question 22 · open in the official exam PDF (p. 13) ↗

Principal reduction for payment 2 in an amortisation table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 23 · 1 mark**

2025 VCAA Exam 1 · Question 21 · open in the official exam PDF (p. 11) ↗

When total perpetuity payments first exceed the amount invested

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 24 · 1 mark**

2025 VCAA Exam 1 · Question 22 · open in the official exam PDF (p. 11) ↗

Interest rate needed for a weekly-compounding savings plan with $50 weekly deposits to reach $14 000 in 3 years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

## Matrices

**Question 25 · 1 mark**

2023 VCAA Exam 1 · Question 25 · open in the official exam PDF (p. 15) ↗

Meaning of element $m_{21}$ in a matrix of daily temperatures

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 26 · 1 mark**

2023 VCAA Exam 1 · Question 26 · open in the official exam PDF (p. 15) ↗

Which three letters change position when a column matrix is multiplied by a permutation matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 27 · 1 mark**

2024 VCAA Exam 1 · Question 25 · open in the official exam PDF (p. 15) ↗

Order of a matrix added to a product of a 2 × 3 and a 3 × 1 matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 28 · 1 mark**

2024 VCAA Exam 1 · Question 28 · open in the official exam PDF (p. 16) ↗

Meaning of element $x_{31}$ after multiplying a sport-by-team matrix by a column of 1s

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 29 · 1 mark**

2025 VCAA Exam 1 · Question 25 · open in the official exam PDF (p. 13) ↗

Which description fits a given 3 × 3 matrix of 0s and 1s

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 30 · 1 mark**

2025 VCAA Exam 1 · Question 26 · open in the official exam PDF (p. 13) ↗

Calculation for element $c_{21}$ of a matrix product

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 31 · 1 mark**

2025 VCAA Exam 1 · Question 28 · open in the official exam PDF (p. 14) ↗

Which sequence passes a message from A to E using a communication matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 32 · 1 mark**

2025 VCAA Exam 1 · Question 29 · open in the official exam PDF (p. 15) ↗

Leslie matrix matching a life-cycle diagram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

## Networks and decision mathematics

**Question 33 · 1 mark**

2023 VCAA Exam 1 · Question 34 · open in the official exam PDF (p. 20) ↗

What a bipartite graph is typically used for

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 34 · 1 mark**

2023 VCAA Exam 1 · Question 36 · open in the official exam PDF (p. 21) ↗

Minimum total time to allocate four duties to four employees

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

**Question 35 · 1 mark**

2024 VCAA Exam 1 · Question 33 · open in the official exam PDF (p. 21) ↗

Sum of the degrees of a small graph with a loop and an isolated vertex

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 36 · 1 mark**

2024 VCAA Exam 1 · Question 34 · open in the official exam PDF (p. 21) ↗

Which route is an Eulerian trail through a graph

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 37 · 1 mark**

2024 VCAA Exam 1 · Question 36 · open in the official exam PDF (p. 23) ↗

Which tree is the minimum spanning tree for cabling eight houses

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 38 · 1 mark**

2025 VCAA Exam 1 · Question 33 · open in the official exam PDF (p. 18) ↗

Number of Hamiltonian cycles starting from E

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 39 · 1 mark**

2025 VCAA Exam 1 · Question 34 · open in the official exam PDF (p. 18) ↗

Number of bridges in a graph

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

**Question 40 · 1 mark**

2025 VCAA Exam 1 · Question 36 · open in the official exam PDF (p. 20) ↗

Capacity of a marked cut in a water pipe network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

## Exam 2 · written response (60 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

### Data analysis

**Question 1 (5 marks)**

2023 VCAA Exam 2 · Question 2 · open in the official exam PDF (p. 4) ↗

Oyster sizes: a frequency table, a percentage segmented bar chart, and a two-way table comparing two farms

**a.i.** Complete the frequency table (numbers) for 20 oysters (1 mark)

**a.ii.** Draw a percentage segmented bar chart from the table (1 mark)

**b.i.** Percentage of all oysters graded large, to the nearest whole number (1 mark)

**b.ii.** Does the two-way table support farm A growing larger oysters? Compare two percentages (2 marks)

**Question 2 (4 marks)**

2023 VCAA Exam 2 · Question 4 · open in the official exam PDF (p. 8) ↗

Monthly ice cream consumption time series over three years: seasonality, deseasonalising and a seasonal index

**a.** A feature of the plot that shows seasonality (1 mark)

**b.** Deseasonalised April 2010 value given a seasonal index of 1.05 (1 mark)

**c.** Show that the July 2011 seasonal index is 1.10 from a year of monthly data (2 marks)

**Question 3 (4 marks)**

2024 VCAA Exam 2 · Question 2 · open in the official exam PDF (p. 5) ↗

Boxplot of women's Olympic high jump gold medal heights (Wgold): shape, counting values and fences

**a.** Describe the shape of the distribution (1 mark)

**b.** Smallest possible number of heights below 1.85 m (1 mark)

**c.i.** Show the lower and upper fences are 1.565 and 2.325 (1 mark)

**c.ii.** Explain why there are no outliers (1 mark)

**Question 4 (2 marks)**

2024 VCAA Exam 2 · Question 4 · open in the official exam PDF (p. 9) ↗

Time series of women's gold medal heights 1952–1988: five-median smoothing and qualitative features

**a.** Complete the five-median smoothing on the plot (1 mark)

**b.** Two qualitative features of the plot (1 mark)

**Question 5 (5 marks)**

2025 VCAA Exam 2 · Question 1 · open in the official exam PDF (p. 2) ↗

Sale prices of 20 homes (apartments and houses): median, variable type, standard deviation and percentages

**a.** Median price of all 20 homes (1 mark)

**b.** Is “type” numerical, nominal or ordinal? (1 mark)

**c.i.** Standard deviation of apartment prices, to the nearest dollar (1 mark)

**c.ii.** Compare the spread of house and apartment prices using the standard deviations (1 mark)

**d.** Complete a table of percentages in price ranges by home type (1 mark)

**Question 6 (2 marks)**

2025 VCAA Exam 2 · Question 2 · open in the official exam PDF (p. 4) ↗

A boxplot of 203 home sale prices: range and upper fence

**a.** Range of the sale prices (1 mark)

**b.** Upper fence (1 mark)

**Question 7 (2 marks)**

2025 VCAA Exam 2 · Question 3 · open in the official exam PDF (p. 5) ↗

Normally distributed sale prices with mean $1 400 000: find σ from a z-score, then a percentage with the 68–95–99.7% rule

Percentage of homes priced between $560 000 and $1 680 000 (2 marks)

## Recursion and financial modelling

**Question 8 (3 marks)**

2023 VCAA Exam 2 · Question 5 · open in the official exam PDF (p. 10) ↗

A $30 000 motorcycle loan at 6.4% p.a. compounding quarterly, repaid over six years

**a.** Total number of repayments (1 mark)

**b.** Balance after two quarters, showing recursive calculations (1 mark)

**c.** Final repayment that differs slightly from the others (1 mark)

**Question 9 (4 marks)**

2024 VCAA Exam 2 · Question 5 · open in the official exam PDF (p. 10) ↗

Grooming equipment depreciating by $V_n = 15\,000 - 60n$ (n weeks)

**a.** Depreciation each week (1 mark)

**b.** Value after four years (1 mark)

**c.** Recurrence relation for the weekly value (1 mark)

**d.** Equivalent annual flat rate percentage (1 mark)

**Question 10 (2 marks)**

2024 VCAA Exam 2 · Question 6 · open in the official exam PDF (p. 11) ↗

Savings compounding fortnightly with an effective rate of 5.07%

**a.** Nominal annual rate to two decimal places (1 mark)

**b.** Why the nominal rate is lower than the effective rate (1 mark)

**Question 11 (3 marks)**

2025 VCAA Exam 2 · Question 8 · open in the official exam PDF (p. 12) ↗

Flat rate depreciation of $40 000 lighting equipment shown on a graph

**a.i.** Recurrence relation for the value (1 mark)

**a.ii.** Rule for $V_n$ in terms of n (1 mark)

**b.** Annual flat rate percentage (1 mark)

## Matrices

**Question 12 (2 marks)**

2023 VCAA Exam 2 · Question 11 · open in the official exam PDF (p. 18) ↗

Circus workers continuing (W) or leaving (L), modelled with $S_{n+1} = TS_n + B$

**a.** Transition matrix T (1 mark)

**b.** Matrix B that keeps the workforce at 180 (1 mark)

**Question 13 (3 marks)**

2024 VCAA Exam 2 · Question 9 · open in the official exam PDF (p. 14) ↗

Hourly pay rates for three shift types in a row matrix R

**a.** Write down $R^T$ (1 mark)

**b.** Matrix calculation for a week of 28 normal, 6 overtime and 8 weekend hours (1 mark)

**c.** Values of n and p when Q = n[1 1.5 2 p] = [36 54 72 90] (1 mark)

**Question 14 (2 marks)**

2024 VCAA Exam 2 · Question 10 · open in the official exam PDF (p. 16) ↗

Security codes built from the rule $(i - j)^2 + 2j$

**a.** The five numbers in the 1 × 5 code matrix (1 mark)

**b.** Last number of an eight-number code with the same rule (1 mark)

**Question 15 (3 marks)**

2025 VCAA Exam 2 · Question 11 · open in the official exam PDF (p. 15) ↗

Room capacities and weekly enrolments at an early learning centre

**a.** Order of the enrolment matrix E (1 mark)

**b.** What [1 1 1] × E gives (1 mark)

**c.** Diagonal matrix F for the new capacities (+25%, +50%, −10%) (1 mark)

**Question 16 (2 marks)**

2025 VCAA Exam 2 · Question 12 · open in the official exam PDF (p. 16) ↗

Children moving between rooms or leaving (absorbing state), then new enrolments each year

**a.** Find the previous year's state $S_{2023}$ from $S_{2024}$ (1 mark)

**b.** Expected total enrolled at the start of 2026 with new children added each year (1 mark)

## Networks and decision mathematics

**Question 17 (4 marks)**

2023 VCAA Exam 2 · Question 12 · open in the official exam PDF (p. 20) ↗

A graph of shared borders between five states: degrees, Euler's formula and matching to a map

**a.** Sum of the degrees of the graph (1 mark)

**b.i.** Fill in Euler's formula for this graph (1 mark)

**b.ii.** Why Euler's formula holds for this graph (1 mark)

**c.** Match states B to E with the numbered regions of a map (1 mark)

**Question 18 (4 marks)**

2024 VCAA Exam 2 · Question 13 · open in the official exam PDF (p. 22) ↗

Supermarket departments as a graph of shared boundaries: adding a vertex, a Hamiltonian path, and a floorplan from an adjacency matrix

**a.** Draw the missing vertex and edges (1 mark)

**b.i.** Finishing department for a journey from Promotional visiting each department once (1 mark)

**b.ii.** Mathematical name for that journey (1 mark)

**c.** Label a new floorplan from an adjacency matrix (1 mark)

**Question 19 (4 marks)**

2025 VCAA Exam 2 · Question 15 · open in the official exam PDF (p. 22) ↗

A graph of roads between Frances's favourite locations

**a.** Sum of the degrees of the vertices (1 mark)

**b.** Complete Euler's formula for this graph (1 mark)

**c.** Name for a route from the gym visiting each other location once, ending at home (1 mark)

**d.** Draw a spanning tree using the graph's edges (1 mark)

# Marking guide

Mark yourself only after you've finished both exams. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

## Exam 1 · multiple choice

| ✓ | Q | Ans | Why | How Victoria went |
|---|---|---|---|---|
| ☐ | 1 | **B** | With 40 values the median is the mean of the 20th and 21st values; counting up the dot plot gives 136 s. | 90% correct<br>A 1% · B 90% · C 6% · D 2% · E 1% |
| ☐ | 2 | **A** | The long tail is on the right and the slowest times are well away from the rest: positively skewed with possible outliers. | 88% correct<br>A 88% · B 4% · C 1% · D 0% · E 7% |
| ☐ | 3 | **E** | Six values with mean 8 total 48. The five given total 39, so day six needs 9. | 89% correct<br>A 1% · B 2% · C 3% · D 4% · E 89% |
| ☐ | 4 | **D** | 60 and 104 minutes are $82 \mp 2 \times 11$, so 95% of 2380 = 2261. | 82% correct<br>A 1% · B 5% · C 7% · D 82% · E 4% |
| ☐ | 5 | **B** | $x = 163.56 - 0.85 \times 8.14 = 156.6$ cm. | 84% correct<br>A 4% · B 84% · C 6% · D 4% · E 1% |
| ☐ | 6 | **A** | Both record a category (cash/electronic; under/over $50), so both are categorical. | 79% correct<br>A 79% · B 3% · C 16% · D 2% |
| ☐ | 7 | **C** | The five-number summary gives a median of 1.5 (the 12th and 13th values are 1 and 2). | 80% correct<br>A 10% · B 4% · C 80% · D 6% |
| ☐ | 8 | **C** | $-1.75 = (48 - 55.7) \div s$, so $s = 7.7 \div 1.75 = 4.4$. | 82% correct<br>A 5% · B 9% · C 82% · D 4% |
| ☐ | 9 | **D** | Read the five-number summaries: T's IQR is smaller, its median is less than 10 years higher, and H's $Q_3$ is below T's $Q_1$. T's minimum is above H's median, so D is correct. | 82% correct<br>A 7% · B 4% · C 7% · D 82% |
| ☐ | 10 | **C** | 2.5% above 178.9 means $178.9 = \mu + 2\sigma$; 16% below 157.6 means $157.6 = \mu - \sigma$. So $3\sigma = 21.3$, $\sigma = 7.1$ and $\mu = 164.7$. | 81% correct<br>A 6% · B 10% · C 81% · D 3% |
| ☐ | 11 | **C** | Lower fence = 74.9 − 1.5 × 3.6 = 69.5. Three values (68.5, 68.6, 69.0) are below it. | 79% correct<br>A 5% · B 10% · C 79% · D 6% |
| ☐ | 12 | **B** | 42 of the 105 female buyers: 40%. | 86% correct<br>A 2% · B 86% · C 6% · D 6% |
| ☐ | 13 | **D** | Two categorical variables: a segmented bar chart. | 79% correct<br>A 9% · B 5% · C 7% · D 79% |
| ☐ | 14 | **C** | Transform life on CAS, then fit $\log_{10}$(life) on doctors: $\log_{10}(\text{life}) = 1.79 + 0.0383 \times$ doctors. | 81% correct<br>A 5% · B 10% · C 81% · D 4% |
| ☐ | 15 | **D** | The margins drift down over the season with irregular ups and downs: decreasing trend with irregular fluctuations. | 85% correct<br>A 13% · B 1% · C 1% · D 85% |
| ☐ | 16 | **B** | Mean of days 6–9 is 134.25 and of days 7–10 is 145.75; centring gives 140.0. | 81% correct<br>A 8% · B 81% · C 6% · D 5% |
| ☐ | 17 | **C** | Subtracting 1314 each year gives $G_n = 15\,000 - 1314n$. | 60% correct<br>A 33% · B 2% · C 60% · D 4% · E 1% |

| | | | | |
|---|---|---|---|---|
| ☐ | 18 | **E** | 1314 ÷ 0.04 = 32 850 cups. | 62% correct<br>A 16% · B 11% · C 4% · D 5% · E 62% |
| ☐ | 19 | **C** | It loses 2400 over 4 years, 600 per year, which is 20% of 3000. | 82% correct<br>A 2% · B 5% · C 82% · D 6% · E 5% |
| ☐ | 20 | **B** | (1.00075 − 1) × 52 × 100 = 3.9% per annum. | 74% correct<br>A 11% · B 74% · C 7% · D 8% |
| ☐ | 21 | **C** | Keep 82% each year: k = 0.82. | 72% correct<br>A 4% · B 13% · C 72% · D 11% |
| ☐ | 22 | **C** | Monthly rate = 960 ÷ 240 000 = 0.4%. Interest = 0.004 × 238 218.95 = 952.88; principal reduction = 2741.05 − 952.88 = $1788.17. | 70% correct<br>A 9% · B 16% · C 70% · D 5% |
| ☐ | 23 | **C** | Each payment is 5% of 250 000 = 12 500, and 250 000 ÷ 12 500 = 20, so the total first exceeds it at the end of year 21. | 69% correct<br>A 8% · B 16% · C 69% · D 6% |
| ☐ | 24 | **A** | N = 156, PV = −4000, Pmt = −50, FV = 14 000, PpY = CpY = 52: I% ≈ 8.4. | 68% correct<br>A 68% · B 13% · C 12% · D 6% |
| ☐ | 25 | **A** | Row 2 is week 2 and column 1 is Monday: 29 °C on Monday in week 2. | 89% correct<br>A 89% · B 6% · C 1% · D 2% · E 2% |
| ☐ | 26 | **B** | PQ reorders the rows: t and s stay in place while e, a and m swap around. | 90% correct<br>A 2% · B 90% · C 4% · D 3% · E 2% |
| ☐ | 27 | **B** | JK is 2 × 1, and only a matrix of the same order can be added to it: 2 × 1. | 79% correct<br>A 9% · B 79% · C 5% · D 7% |
| ☐ | 28 | **C** | Multiplying by a column of 1s adds across each row. Row 3 is tennis: 63 + 76 + 66 + 75 = 280 students compete in tennis. | 84% correct<br>A 3% · B 2% · C 84% · D 11% |
| ☐ | 29 | **A** | Every element is 0 or 1: binary. It isn't a permutation matrix (row 2 has two 1s and row 3 has none). | 83% correct<br>A 83% · B 9% · C 5% · D 3% |
| ☐ | 30 | **C** | Row 2 of A times column 1 of B: 1 × 3 + 6 × 5. | 84% correct<br>A 8% · B 3% · C 84% · D 4% |
| ☐ | 31 | **C** | Check each link in the matrix: A→B, B→D and D→E all exist. Each other option has a missing link. | 93% correct<br>A 1% · B 4% · C 93% · D 2% |
| ☐ | 32 | **A** | Birth rates (0, 2.1, 4.6, 1.8) go in row 1, and survival rates 0.9, 0.7, 0.2 just below the leading diagonal: matrix A. | 84% correct<br>A 84% · B 9% · C 5% · D 1% |
| ☐ | 33 | **A** | Bipartite graphs model allocation: matching tasks to workers. | 77% correct<br>A 77% · B 5% · C 7% · D 6% · E 4% |
| ☐ | 34 | **B** | The Hungarian algorithm (or checking allocations) gives a minimum of 30 minutes. | 75% correct<br>A 10% · B 75% · C 6% · D 5% · E 2% |
| ☐ | 35 | **C** | 1 + 3 + 4 + 2 + 2 + 0 = 12 (the loop adds 2 to its vertex). | 75% correct<br>A 6% · B 17% · C 75% · D 2% |
| ☐ | 36 | **C** | An Eulerian trail starts and ends at the two odd vertices and uses every edge exactly once: BACBDCFDEF. | 85% correct<br>A 7% · B 4% · C 85% · D 3% |
| ☐ | 37 | **D** | Apply Prim's algorithm from any house, always taking the shortest cable to a new house: tree D. | 83% correct<br>A 4% · B 9% · C 4% · D 83% |
| ☐ | 38 | **B** | Only E–D–C–B–A–F–E and its reverse E–F–A–B–C–D–E: 2. | 76% correct<br>A 7% · B 76% · C 6% · D 11% |

| | | | | |
|---|---|---|---|---|
| ☐ | 39 | **C** | Removing any one of the three top edges disconnects the graph: 3 bridges. | 85% correct<br>A 8% · B 4% · C 85% · D 3% |
| ☐ | 40 | **A** | Cut 1 crosses six pipes; only the four flowing from source side to sink side count: 7 + 3 + 5 + 12 = 27. | 77% correct<br>A 77% · B 12% · C 3% · D 7% |

**Multiple-choice score:** ☐ **/ 40** **(state average 32.0)**

## Exam 2 · written response

### Question 1 (5 marks) · state average 4 / 5 (80%)

2023 VCAA Exam 2 · Question 2

**a.i.** Complete the frequency table (numbers) for 20 oysters

☐ small 7, medium 10, large 3, total 20

**Answer / notes:** Small 7, medium 10, large 3, total 20. Tally marks or a missing total weren't accepted.

State average 1.0 / 1 (100%); 95% scored full marks. Mark distribution: 0: 5% · 1: 95%

**Your mark:** ☐ **/ 1**

**a.ii.** Draw a percentage segmented bar chart from the table

☐ Segments at 35%, 50% and 15% (boundaries at 35 and 85), matching the key

**Answer / notes:** One bar split at 35% and 85%. Well done.

State average 0.8 / 1 (80%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

**Your mark:** ☐ **/ 1**

**b.i.** Percentage of all oysters graded large, to the nearest whole number

☐ 17%

**Answer / notes:** (44 + 46) ÷ 530 ≈ 17%.

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

**Your mark:** ☐ **/ 1**

**b.ii.** Does the two-way table support farm A growing larger oysters? Compare two percentages

☐ Large at farm A 21% and farm B 14%

☐ Yes, because farm A's percentage of large oysters is higher

**Answer / notes:** Yes: 44 ÷ 210 ≈ 21% of farm A's oysters are large vs 46 ÷ 320 ≈ 14% for farm B. You had to state the comparison explicitly.

State average 1.3 / 2 (65%); 50% scored full marks. Mark distribution: 0: 19% · 1: 32% · 2: 50%

**Your mark:** ☐ **/ 2**

### Question 2 (4 marks) · state average 1.7 / 4 (43%)

2023 VCAA Exam 2 · Question 4

**a.** A feature of the plot that shows seasonality

☐ Peaks (or troughs) occur 12 months apart

**Answer / notes:** The maximum values occur 12 months apart. Writing a definition of seasonality didn't score.

State average 0.3 / 1 (30%); 26% scored full marks. Mark distribution: 0: 74% · 1: 26%

**Your mark:** ☐ **/ 1**

**b.** Deseasonalised April 2010 value given a seasonal index of 1.05

☐ 0.17 L/person

**Answer / notes:** Actual April value ÷ 1.05 ≈ 0.17.

State average 0.7 / 1 (70%); 72% scored full marks. Mark distribution: 0: 28% · 1: 72%

**Your mark:** ☐ **/ 1**

**c.** Show that the July 2011 seasonal index is 1.10 from a year of monthly data

- [ ] Monthly mean for 2011 ≈ 0.167
- [ ] Index = 0.183 ÷ 0.16700 ≈ 1.0958 ≈ 1.10

**Answer / notes:** Mean of the 12 months ≈ 0.167; 0.183 ÷ 0.167 ≈ 1.096 = 1.10. Both calculations were needed.

State average 0.7 / 2 (35%); 21% scored full marks. Mark distribution: 0: 50% · 1: 29% · 2: 21%

**Your mark:** ☐ **/ 2**

### Question 3 (4 marks) · state average 2.7 / 4 (68%)

2024 VCAA Exam 2 · Question 2

**a.** Describe the shape of the distribution

- [ ] negatively skewed

**Answer / notes:** Negatively skewed (no outliers was also accepted).

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

**Your mark:** ☐ **/ 1**

**b.** Smallest possible number of heights below 1.85 m

- [ ] 1

**Answer / notes:** Only one value is needed to stretch the whisker below $Q_1 = 1.85$. Most wrote a height (1.67) instead of a count; just 3% were right.

State average 0.1 / 1 (10%); 3% scored full marks. Mark distribution: 0: 97% · 1: 3%

**Your mark:** ☐ **/ 1**

**c.i.** Show the lower and upper fences are 1.565 and 2.325

- [ ] IQR = 2.04 − 1.85 = 0.19, so fences 1.85 − 1.5 × 0.19 = 1.565 and 2.04 + 1.5 × 0.19 = 2.325

**Answer / notes:** Show the IQR and both fence calculations.

State average 0.9 / 1 (90%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

**Your mark:** ☐ **/ 1**

**c.ii.** Explain why there are no outliers

- [ ] The minimum (1.67) is above 1.565 and the maximum (2.06) is below 2.325

**Answer / notes:** Compare the minimum with the lower fence and the maximum with the upper fence separately.

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

**Your mark:** ☐ **/ 1**

### Question 4 (2 marks) · state average 0.9 / 2 (45%)

2024 VCAA Exam 2 · Question 4

**a.** Complete the five-median smoothing on the plot

- [ ] All smoothed points correctly placed and joined with a dashed line

**Answer / notes:** For each year, plot the median of the five values centred on it.

State average 0.6 / 1 (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

**Your mark:** ☐ **/ 1**

**b.** Two qualitative features of the plot

- [ ] increasing trend and irregular fluctuations

**Answer / notes:** Increasing trend with irregular fluctuations. There's no seasonality, outlier or structural change; use study-design terms.

State average 0.3 / 1 (30%); 24% scored full marks. Mark distribution: 0: 76% · 1: 24%

**Your mark:** ☐ **/ 1**

**Question 5 (5 marks) · state average 3.8 / 5 (76%)**

2025 VCAA Exam 2 · Question 1

**a.** Median price of all 20 homes

☐ $920 000

**Answer / notes:** The 10th and 11th values are both 920 000.

State average 0.9 / 1 (90%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

**Your mark:** ☐ **/ 1**

**b.** Is “type” numerical, nominal or ordinal?

☐ nominal

**Answer / notes:** Nominal (apartment/house has no order).

State average 0.8 / 1 (80%); 77% scored full marks. Mark distribution: 0: 23% · 1: 77%

**Your mark:** ☐ **/ 1**

**c.i.** Standard deviation of apartment prices, to the nearest dollar

☐ $346 466

**Answer / notes:** $346 466. Enter the data carefully.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

**Your mark:** ☐ **/ 1**

**c.ii.** Compare the spread of house and apartment prices using the standard deviations

☐ Houses have a lower spread than apartments (smaller standard deviation)

**Answer / notes:** Use the standard deviations given; range or IQR weren’t appropriate here.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

**Your mark:** ☐ **/ 1**

**d.** Complete a table of percentages in price ranges by home type

☐ House 0, 40, 60; apartment 30, 50, 20

**Answer / notes:** Houses: 0% under $600 000, 40% from $600 000 to $1 000 000, 60% over. Apartments: 30%, 50%, 20%.

State average 0.9 / 1 (90%); 89% scored full marks. Mark distribution: 0: 11% · 1: 89%

**Your mark:** ☐ **/ 1**

**Question 6 (2 marks) · state average 1.7 / 2 (85%)**

2025 VCAA Exam 2 · Question 2

**a.** Range of the sale prices

☐ $900 000

**Answer / notes:** 1 300 000 − 400 000 = $900 000.

State average 0.9 / 1 (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

**Your mark:** ☐ **/ 1**

**b.** Upper fence

☐ $1 350 000

**Answer / notes:** 900 000 + 1.5 × (900 000 − 600 000) = $1 350 000.

State average 0.8 / 1 (80%); 82% scored full marks. Mark distribution: 0: 18% · 1: 82%

**Your mark:** ☐ **/ 1**

**Question 7 (2 marks) · state average 1.1 / 2 (55%)**

2025 VCAA Exam 2 · Question 3

Percentage of homes priced between $560 000 and $1 680 000

- [ ] σ = (1 400 000 − 952 000) ÷ 1.6 = 280 000
- [ ] 560 000 is μ − 3σ and 1 680 000 is μ + σ: 49.85% + 34% = 83.85%

**Answer / notes:** 83.85%. Most found σ; many then lost the second mark by rounding when no rounding was asked for.

State average 1.1 / 2 (55%); 47% scored full marks. Mark distribution: 0: 41% · 1: 13% · 2: 47%

**Your mark:** ____ **/ 2**

### Question 8 (3 marks) · state average 1.6 / 3 (53%)

2023 VCAA Exam 2 · Question 5

**a.** Total number of repayments

- [ ] 24

**Answer / notes:** 6 × 4 = 24.

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

**Your mark:** ____ **/ 1**

**b.** Balance after two quarters, showing recursive calculations

- [ ] $A_1 = 1.016 \times 30\,000 - 1515.18 = 28\,964.82$ and $A_2 = 1.016 \times 28\,964.82 - 1515.18 = 27\,913.08$

**Answer / notes:** $A_1 = 28\,964.82$, $A_2 = 27\,913.08$. Marks were lost for not showing the steps, rounding early or transcription errors.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

**Your mark:** ____ **/ 1**

**c.** Final repayment that differs slightly from the others

- [ ] $1515.04

**Answer / notes:** After 23 repayments, the balance grown by one quarter's interest is $1515.04. Some added the small leftover instead of subtracting it.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

**Your mark:** ____ **/ 1**

### Question 9 (4 marks) · state average 2.6 / 4 (65%)

2024 VCAA Exam 2 · Question 5

**a.** Depreciation each week

- [ ] $60

**Answer / notes:** $60 per week.

State average 0.9 / 1 (90%); 87% scored full marks. Mark distribution: 0: 13% · 1: 87%

**Your mark:** ____ **/ 1**

**b.** Value after four years

- [ ] $2520

**Answer / notes:** 15 000 − 60 × 208 = $2520. $14 760 forgot 52 weeks per year.

State average 0.7 / 1 (70%); 72% scored full marks. Mark distribution: 0: 28% · 1: 72%

**Your mark:** ____ **/ 1**

**c.** Recurrence relation for the weekly value

- [ ] $V_0 = 15\,000$, $V_{n+1} = V_n - 60$

**Answer / notes:** $V_0 = 15\,000$, $V_{n+1} = V_n - 60$. Don't add n to the end.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

**Your mark:** ____ **/ 1**

**d.** Equivalent annual flat rate percentage

☐ 20.8%

**Answer / notes:** 60 × 52 = 3120 per year; 3120 ÷ 15 000 = 20.8%. 0.4% (weekly) was a common error.

State average 0.4 / 1 (40%); 42% scored full marks. Mark distribution: 0: 58% · 1: 42%

**Your mark:** ☐ **/ 1**

### Question 10 (2 marks) · state average 0.8 / 2 (40%)

2024 VCAA Exam 2 · Question 6

**a.** Nominal annual rate to two decimal places

☐ 4.95%

**Answer / notes:** Solve $[(1 + r/2600)^{26} - 1] \times 100 = 5.07$: $r \approx 4.95\%$.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 50% · 1: 51%

**Your mark:** ☐ **/ 1**

**b.** Why the nominal rate is lower than the effective rate

☐ The nominal rate doesn't include the effect of fortnightly compounding

**Answer / notes:** The effective rate includes interest earned on interest during the year; the nominal rate doesn't. Many could define the terms but not explain this.

State average 0.3 / 1 (30%); 26% scored full marks. Mark distribution: 0: 74% · 1: 26%

**Your mark:** ☐ **/ 1**

### Question 11 (3 marks) · state average 2 / 3 (67%)

2025 VCAA Exam 2 · Question 8

**a.i.** Recurrence relation for the value

☐ $V_0 = 40\,000$, $V_{n+1} = V_n - 8000$

**Answer / notes:** It falls by 8000 each year.

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

**Your mark:** ☐ **/ 1**

**a.ii.** Rule for $V_n$ in terms of n

☐ $V_n = 40\,000 - 8000n$

**Answer / notes:** A rule gives $V_n$ directly; many repeated the recurrence relation.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

**Your mark:** ☐ **/ 1**

**b.** Annual flat rate percentage

☐ 20%

**Answer / notes:** 8000 ÷ 40 000 = 20%.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

**Your mark:** ☐ **/ 1**

### Question 12 (2 marks) · state average 0.7 / 2 (35%)

2023 VCAA Exam 2 · Question 11

**a.** Transition matrix T

☐ T = [0.95 0; 0.05 1]

**Answer / notes:** Column W: 0.95 stay, 0.05 leave. Column L: 0, 1 (they never return). 0.5 for 0.05 and misplaced 1s were common errors.

State average 0.3 / 1 (30%); 34% scored full marks. Mark distribution: 0: 66% · 1: 34%

**Your mark:** ☐ **/ 1**

**b.** Matrix B that keeps the workforce at 180

☐ B = [9; 0] (or [9; −9])

**Answer / notes:** Each show 5% of 180 = 9 leave, so add 9 workers to W.

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

**Your mark:** ☐ **/ 1**

### Question 13 (3 marks) · state average 2.4 / 3 (80%)

2024 VCAA Exam 2 · Question 9

**a.** Write down $R^T$

☐ Column matrix 36, 54, 72

**Answer / notes:** $R^T$ is the 3 × 1 column matrix with 36, 54, 72.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

**Your mark:** ☐ **/ 1**

**b.** Matrix calculation for a week of 28 normal, 6 overtime and 8 weekend hours

☐ $[28\ 6\ 8] \times R^T = [1908]$

**Answer / notes:** 28 × 36 + 6 × 54 + 8 × 72 = 1908.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

**Your mark:** ☐ **/ 1**

**c.** Values of n and p when Q = n[1 1.5 2 p] = [36 54 72 90]

☐ n = 36, p = 2.5

**Answer / notes:** n = 36 and 36p = 90, so p = 2.5.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

**Your mark:** ☐ **/ 1**

### Question 14 (2 marks) · state average 1.1 / 2 (55%)

2024 VCAA Exam 2 · Question 10

**a.** The five numbers in the 1 × 5 code matrix

☐ [2 5 10 17 26]

**Answer / notes:** With i = 1: j = 1 to 5 gives 2, 5, 10, 17, 26. Many mixed up rows and columns.

State average 0.6 / 1 (60%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

**Your mark:** ☐ **/ 1**

**b.** Last number of an eight-number code with the same rule

☐ 65

**Answer / notes:** j = 8: $(1 - 8)^2 + 16 = 65$.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

**Your mark:** ☐ **/ 1**

### Question 15 (3 marks) · state average 2.1 / 3 (70%)

2025 VCAA Exam 2 · Question 11

**a.** Order of the enrolment matrix E

☐ 3 × 5

**Answer / notes:** 3 rows (rooms) × 5 columns (days).

State average 1.0 / 1 (100%); 96% scored full marks. Mark distribution: 0: 4% · 1: 96%

**Your mark:** ☐ **/ 1**

**b.** What [1 1 1] × E gives

- [ ] Total enrolments on each day of the week

**Answer / notes:** Summing down each column gives the daily totals; many described row sums.

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

**Your mark:** ____ **/ 1**

**c.** Diagonal matrix F for the new capacities (+25%, +50%, −10%)

- [ ] F = diag(1.25, 1.5, 0.9)

**Answer / notes:** A 3 × 3 diagonal matrix with 1.25, 1.5 and 0.9 on the leading diagonal.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

**Your mark:** ____ **/ 1**

### Question 16 (2 marks) · state average 0.8 / 2 (40%)

2025 VCAA Exam 2 · Question 12

**a.** Find the previous year's state $S_{2023}$ from $S_{2024}$

- [ ] $S_{2023} = M^{-1}S_{2024} = (16, 20, 25, 0)$

**Answer / notes:** Use the inverse: $S_{2023} = M^{-1}S_{2024}$.

State average 0.6 / 1 (60%); 53% scored full marks. Mark distribution: 0: 47% · 1: 53%

**Your mark:** ____ **/ 1**

**b.** Expected total enrolled at the start of 2026 with new children added each year

- [ ] 50

**Answer / notes:** $S_{2025} = MS_{2024} + B$, $S_{2026} = MS_{2025} + B$; add N, T and P: 15.25 + 15.94 + 19.12 ≈ 50. 115 included children who had left.

State average 0.2 / 1 (20%); 13% scored full marks. Mark distribution: 0: 87% · 1: 13%

**Your mark:** ____ **/ 1**

### Question 17 (4 marks) · state average 3.2 / 4 (80%)

2023 VCAA Exam 2 · Question 12

**a.** Sum of the degrees of the graph

- [ ] 14

**Answer / notes:** 14 (twice the 7 edges).

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

**Your mark:** ____ **/ 1**

**b.i.** Fill in Euler's formula for this graph

- [ ] 5 + 4 = 7 + 2

**Answer / notes:** v = 5, f = 4, e = 7.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

**Your mark:** ____ **/ 1**

**b.ii.** Why Euler's formula holds for this graph

- [ ] planar

**Answer / notes:** It is connected and planar. "Complete" was a common wrong answer.

State average 0.8 / 1 (80%); 81% scored full marks. Mark distribution: 0: 19% · 1: 81%

**Your mark:** ____ **/ 1**

**c.** Match states B to E with the numbered regions of a map

- [ ] B 3, C 2, D 4, E 1

**Answer / notes:** Use each state's neighbours from the graph to place it on the map.

State average 0.7 / 1 (70%); 71% scored full marks. Mark distribution: 0: 29% · 1: 71%

**Your mark:** ____ **/ 1**

### Question 18 (4 marks) · state average 3.4 / 4 (85%)

2024 VCAA Exam 2 · Question 13

**a.** Draw the missing vertex and edges

- [ ] Vertex P with its correct edges

**Answer / notes:** Add Promotional (P) joined to each department it borders.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

**Your mark:** ____ **/ 1**

**b.i.** Finishing department for a journey from Promotional visiting each department once

- [ ] Bakery

**Answer / notes:** Bakery.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

**Your mark:** ____ **/ 1**

**b.ii.** Mathematical name for that journey

- [ ] Hamiltonian path

**Answer / notes:** Hamiltonian path. Writing just "path" didn't score.

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

**Your mark:** ____ **/ 1**

**c.** Label a new floorplan from an adjacency matrix

- [ ] All departments correctly placed

**Answer / notes:** Use each department's boundaries in the matrix to place it.

State average 0.8 / 1 (80%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

**Your mark:** ____ **/ 1**

### Question 19 (4 marks) · state average 3.2 / 4 (80%)

2025 VCAA Exam 2 · Question 15

**a.** Sum of the degrees of the vertices

- [ ] 14

**Answer / notes:** 2 + 4 + 2 + 3 + 3 = 14.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

**Your mark:** ____ **/ 1**

**b.** Complete Euler's formula for this graph

- [ ] 5 + 4 = 7 + 2

**Answer / notes:** v = 5, f = 4, e = 7.

State average 0.9 / 1 (90%); 91% scored full marks. Mark distribution: 0: 9% · 1: 91%

**Your mark:** ____ **/ 1**

**c.** Name for a route from the gym visiting each other location once, ending at home

- [ ] Hamiltonian path

**Answer / notes:** Hamiltonian path.

State average 0.6 / 1 (60%); 53% scored full marks. Mark distribution: 0: 47% · 1: 53%

**Your mark:** ____ **/ 1**

**d.** Draw a spanning tree using the graph's edges

- [ ] Four edges from the original graph connecting all five vertices, no cycles

**Answer / notes:** Many trees work; don't add edges that aren't in the graph.

State average 0.8 / 1 (80%); 81% scored full marks. Mark distribution: 0: 19% · 1: 81%

**Your mark:** ____ **/ 1**

**Exam 1:** ____ **/ 40** **Exam 2:** ____ **/ 60** **Total:** ____ **/ 100**

State average on these exact questions: Exam 1 32.0 / 40, Exam 2 39.8 / 60, total 71.8 / 100 (72%). The notes paraphrase the VCAA examiners' reports; read the full reports for sample answers.